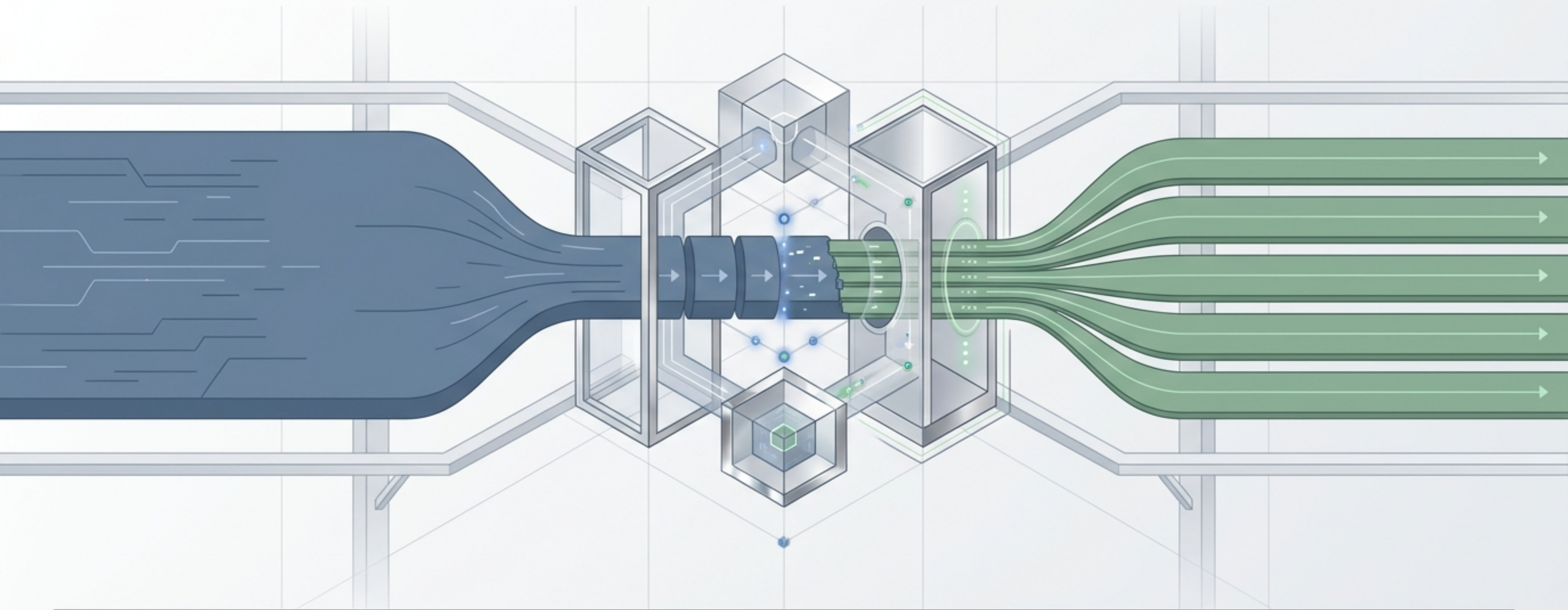


Distilling Digital Intelligence

The AUDSRF Pipeline: Automating Deep Web Discovery & Semantic Filtration

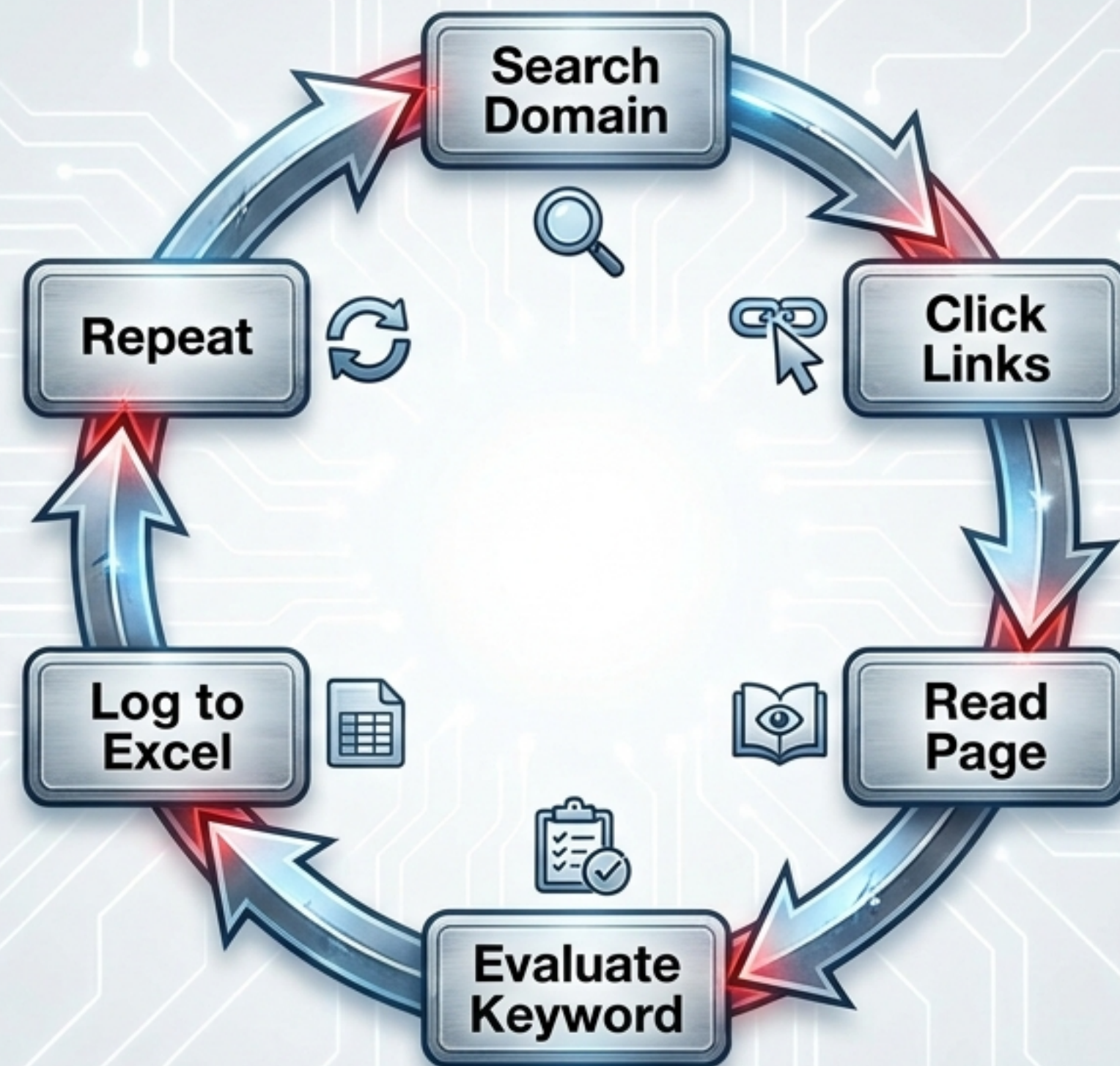


The Dilemma



Organizations possess vast lists of domain-level data, but lack visibility into the specific deep pages that drive actual competitive intelligence.

The Manual Pain



Manual URL discovery is slow, inconsistent, and mathematically impossible to scale across hundreds of noisy domains.



ZONE 01: The Raw Material (Intake & Normalization)

Transforming unstructured, noisy CSVs into a structured, crawler-ready queue.

Pipeline Scope:

Covers Stage 01 (Load & Clean) and Stage 02 (Domain Pre-processing).

Raw Inputs

input_domains.csv

Domain
stadinraksat.fi
rodokselle.fi
autotjaliikenne.fi
vinkkaajaveikkaa.fi
casinopelitnetissa.com
companionlink.com/blog

target_keywords.csv

Topic
Lucky Neko
Mahjong Ways
Chilli Heat Megaways
Extra Juicy Megaways
Floating Dragon Hold Spin Megaways
Madame Destiny Megaways

Normalization Stack

Strip Protocols
(http/https)

Remove Trailing
Slashes

Deduplicate Data

Validate Keywords

Pre-processing Heuristics

✓ Domain Reachability
Check

✓ CMS Pattern
Detection

✓ Homepage Keyword
Density Score

Crawler-Ready
Queue

Pipeline Stage: Load & Clean -> Domain Pre-processing



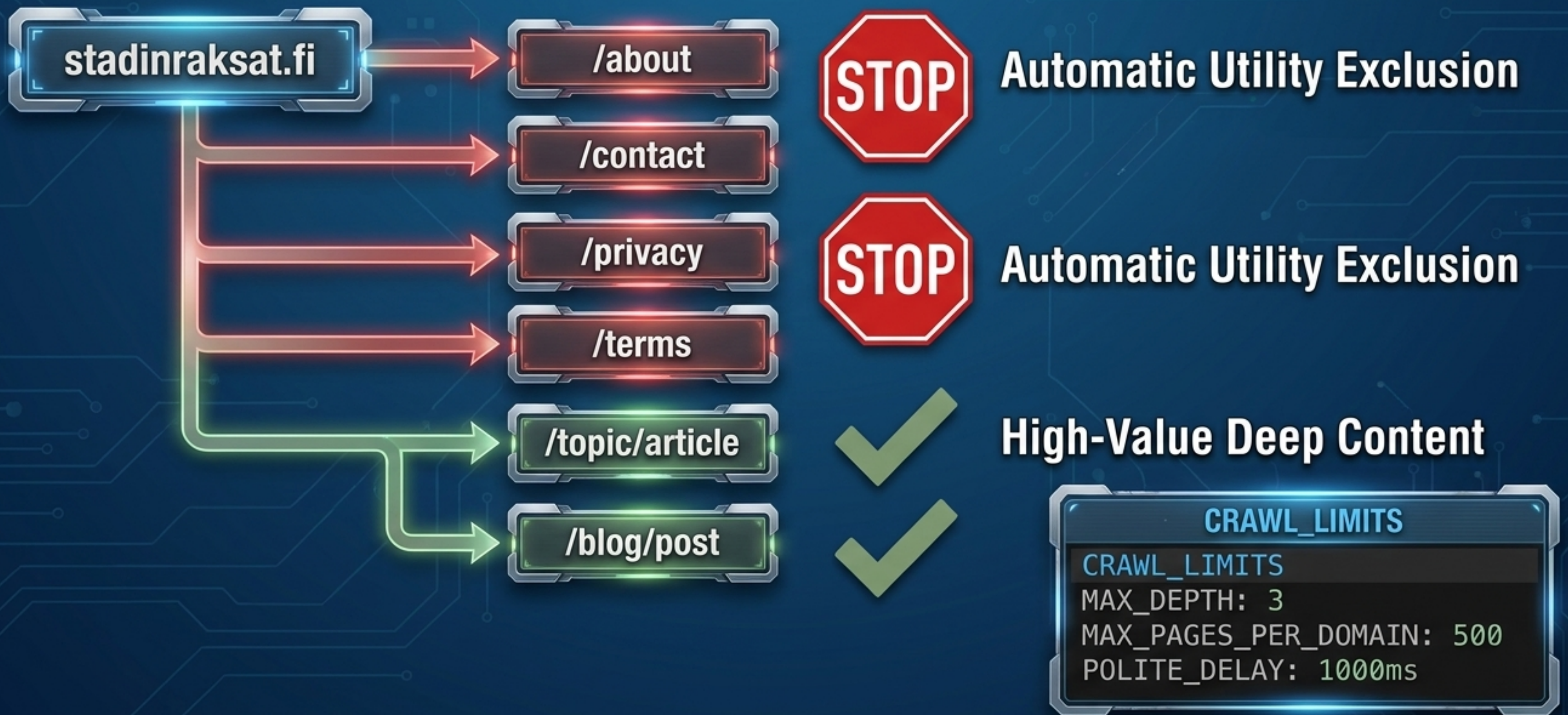
ZONE 02: Deep Exploration (Site Expansion)

Navigating from root domains to granular,
internal page architectures.

Pipeline Scope:

Covers Stage 03 (Queue Initialization) and
Stage 04 (Crawling & Deep URL Discovery).

CRAWLING LOGIC & EXCLUSIONS



Ensuring high performance without infinite loops.



ZONE 03: The Filtration Engine (Semantic Relevance)

Evaluating discovered pages using hybrid lexical and semantic similarity algorithms.

Pipeline Scope:

Covers Stage 05 (Semantic Relevance Filtering).

RELEVANCE SCORING: HYBRID APPROACH

Lexical Match

Mechanism:

Direct keyword matches in Title, H1, URL slug, and body.

Profile:

Deterministic, strong explicit signal. However, fails if a synonym is used.

Semantic Similarity

Mechanism:

TF-IDF vectorization and cosine distance.

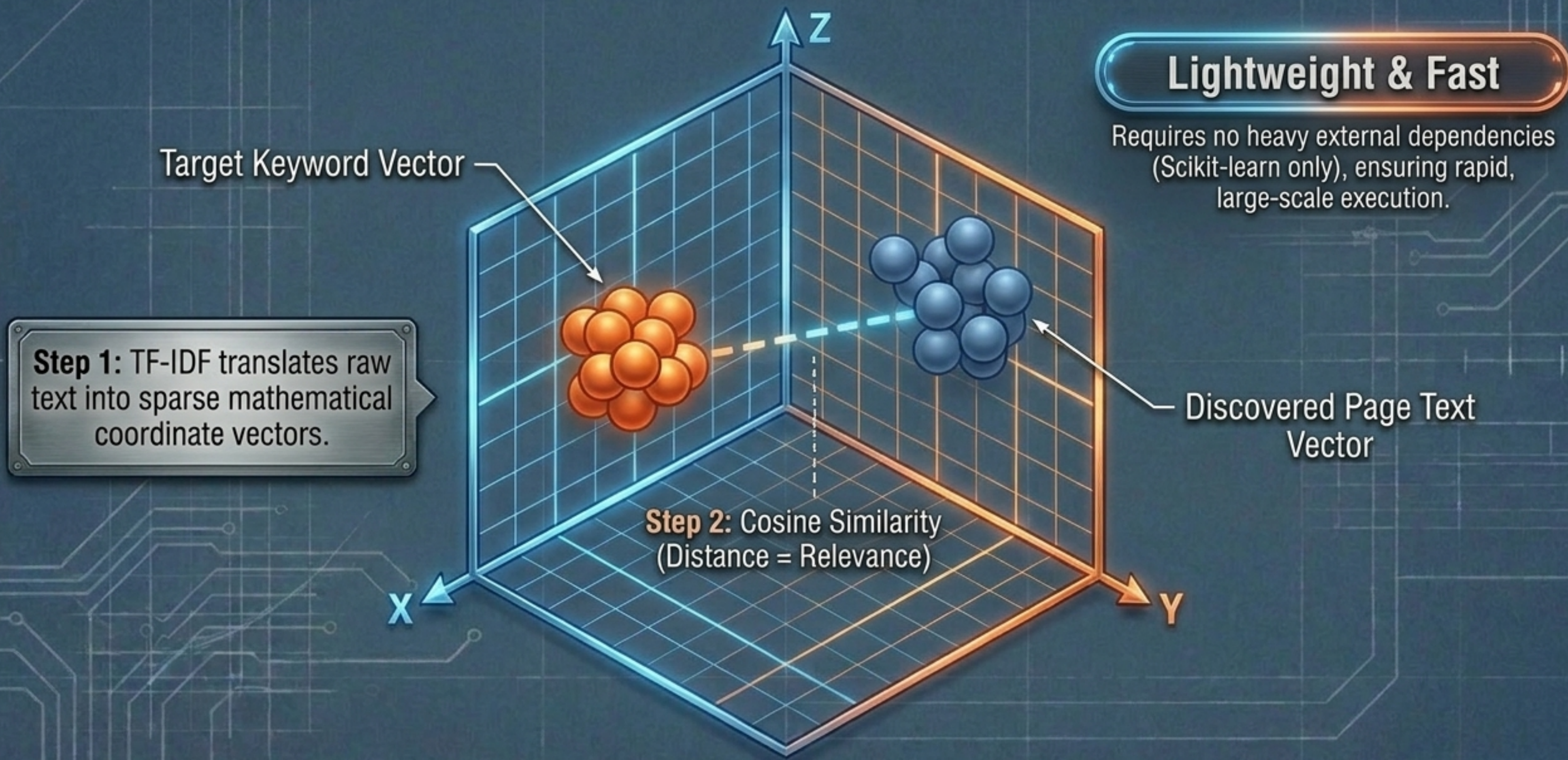
Profile:

Captures softer contextual relationships. However, can produce false positives on loosely related pages.

Hybrid Relevance Scoring

Lexical matches anchor the score; semantic similarity refines it. This reduces false positives and ensures mathematically stable, trustworthy rankings.

SEMANTIC RELEVANCE: VECTORIZATION & COSINE SIMILARITY





ZONE 04: The Refined **Output** (Assembly & QA)

Structuring the intelligence into clean, decision-ready data formats.

Pipeline Scope:

Covers Stage 06 (Assembling Output), Stage 07 (CSV Generation), and Stage 08 (Quality Checks).

<https://addis-techblog.de/it/smarthome>
<https://internetblogger.de/2026/05/jumia-technologies-akte-wkn-a2pgzm-warum-die-aktie-nach-dem-kurssprung-wieder-gefallen-ist>
<https://internetblogger.de/2026/05/ebay-aktualisiert-ds-zahlungsbedingungen>
<https://autotjالييكننه.fi/tahtiautot-esittelyssa-unohtumattomimmat-tvsta-tutut-autot>
<https://companionlink.com/goldmine>
<https://deutsche-staedte.de/bergstrasse/weinbaugebiet.php?start=5>

Source Domain	Target URL	Detected Keyword	Relevance Score
addis-techblog.de	https://addis-techblog.de/it/smarthome	home	0.16
addis-techblog.de	https://addis-techblog.de/it/hardware	war	0.16
addis-techblog.de	https://addis-techblog.de/it/software	war	0.16
internetblogger.de	https://internetblogger.de/2026/05/jumia-technologies-akte-wkn-a2pgzm-warum-die-aktie-nach-dem-kurssprung-wieder-gefallen-ist	war	0.16
internetblogger.de	https://internetblogger.de/2026/05/ebay-aktualisiert-die-zahlungsbedingungen-das-musst-du-wissen	ahl	0.16
autotjالييكننه.fi	https://autotjالييكننه.fi/tahtiautot-esittelyssa-unohtumattomimmat-tvsta-tutut-autot	mma	0.16
autotjالييكننه.fi	https://autotjالييكننه.fi/live-vedonlyonti-ja-moottoriurheilu	live	0.16
companionlink.com	https://companionlink.com/goldmine	gold	0.16
deutsche-staedte.de	https://deutsche-staedte.de/bergstrasse/weinbaugebiet.php?start=5	nba	0.16

Empty inputs handled gracefully. Results ranked strictly by hybrid score.
 Timestamped for precise versioning.

Precision Engine

1.0
Precision Score

Aggressively filtering irrelevant domains and off-topic utility pages.

Granularity Ratio

Deep URLs
Discovered

3

Root
Homepages

1

Successfully identifies high-value internal content over shallow homepages.

Anomaly Detection



Suspicious pages with borderline relevance are automatically isolated for human review to guarantee dataset integrity.



ZONE 05: Next-Gen Upgrades (The LLM Future)

Evolving from statistical keyword matching to deep content intelligence.

Pipeline Scope:

Roadmap and Future
Architecture.

Step 1: TF-IDF (Current State)

Surface Level Semantics

- Relies on exact term overlap.
- Blind to synonyms and contextual intent.

Step 2: Semantic Embeddings

Deep Meaning Capture

- Integrating MiniLM or OpenAI embeddings.
- Detects relevance even when keywords are expressed indirectly.

Step 3: Intent & Interpretation

True Content Intelligence

- LLM-powered classification.
- Identifies user intent, extracts structured insights, and categorizes industry offerings.



High Noise Tolerance

Built specifically for messy, unstructured real-world inputs.

Hybrid Precision

Lexical and Semantic filtering combined to ensure total data trust.

Extensible Architecture

Stage-based design allows modular upgrades (like LLMs) without breaking the core pipeline.

Transforming chaotic web data into structured competitive intelligence—automatically, reliably, and at scale.